

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2021 P2H Worked Solutions

Nov 2021 | P2H

25 questions ■ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

Assistant Lecturer, Cairo University Faculty of Engineering

WhatsApp: +20 112 000 9622 | eliteigcse.com

PAST PAPER QUESTION**Q#: 1****Marks: 4****TOPIC: Algebraic Roots & Indices**

Nov 2021 P2H - Nov 2021 | P2H

- 1 (a) Write down the value of m , given that $3^4 \times 3^5 = 3^m$

$m = \dots\dots\dots$
(1)

- (b) Write down the value of n , given that $(5^3)^7 = 5^n$

$n = \dots\dots\dots$
(1)

- (c) Find the value of p , given that $\frac{7^8 \times 7^2}{7^p} = 7^6$

$p = \dots\dots\dots$
(2)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 4

TOPIC: **Algebraic Roots & Indices**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use index laws(a) $3^4 \times 3^5 = 3^9$, so $m = 9$.**2. Use index laws**(b) $(5^3)^7 = 5^{21}$, so $n = 21$.**3. (c)**

(c)

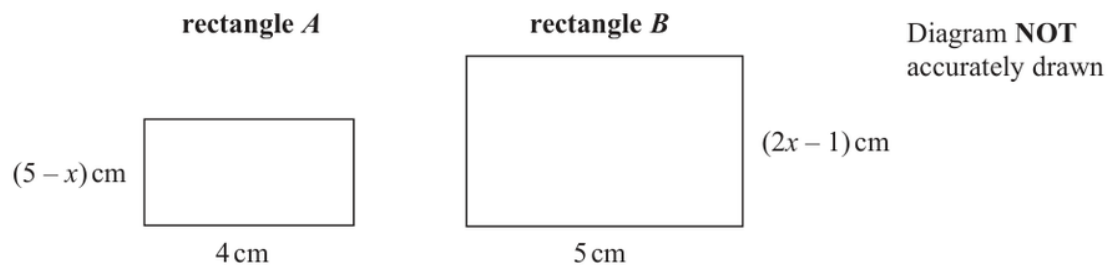
$$\frac{7^8 \times 7^2}{7^p} = 7^{10-p} = 7^6$$

4. Use index lawsSo $10 - p = 6$, hence $p = 4$.**FINAL ANSWER** $m = 9, n = 21, p = 4$

PAST PAPER QUESTION**Q#: 2****Marks: 4****TOPIC: Forming & Solving Equations**

Nov 2021 P2H - Nov 2021 | P2H

- 2 Here are two rectangles, rectangle A and rectangle B .



The area of rectangle B is twice the area of rectangle A .

Work out the value of x .

Show your working clearly.

$x = \dots\dots\dots$

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 4****TOPIC: Forming & Solving Equations**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Area of rectangle A**

Area of rectangle A:

$$4(5 - x)$$

2. Area of rectangle B

Area of rectangle B:

$$5(2x - 1)$$

3. Rectangle B has twice the

Rectangle B has twice the area of rectangle A:

$$5(2x - 1) = 2 \cdot 4(5 - x)$$

$$10x - 5 = 40 - 8x$$

$$18x = 45$$

$$x = 2.5$$

FINAL ANSWER

$$x = 2.5.$$

PAST PAPER QUESTION**Q#: 3****Marks: 5****TOPIC: Statistics Toolkit**

Nov 2021 P2H - Nov 2021 | P2H

- 3** The table gives information about the amounts of money, in euros, that 70 of Anjali's friends spent last Saturday.

Money spent (S euros)	Frequency
$0 < S \leq 8$	6
$8 < S \leq 16$	14
$16 < S \leq 24$	19
$24 < S \leq 32$	25
$32 < S \leq 40$	6

One of Anjali's 70 friends is going to be chosen at random.

- (a) Find the probability that this friend spent more than 24 euros last Saturday.

.....
(1)

- (b) Work out an estimate for the mean amount of money spent by Anjali's friends last Saturday.
Give your answer correct to 2 decimal places.

..... euros
(4)

(Total for Question 3 is 5 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 5****TOPIC: Statistics Toolkit**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. More than euros**

More than 24 euros:

$$25 + 6 = 31$$

2. Total frequency

Total frequency:

$$6 + 14 + 19 + 25 + 6 = 70$$

$$P(S > 24) = \frac{31}{70}$$

3. Use midpoints

Use midpoints 4, 12, 20, 28, 36.

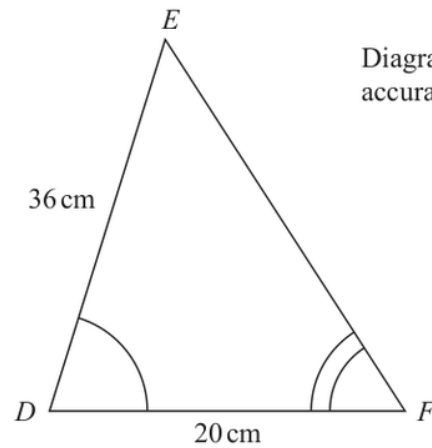
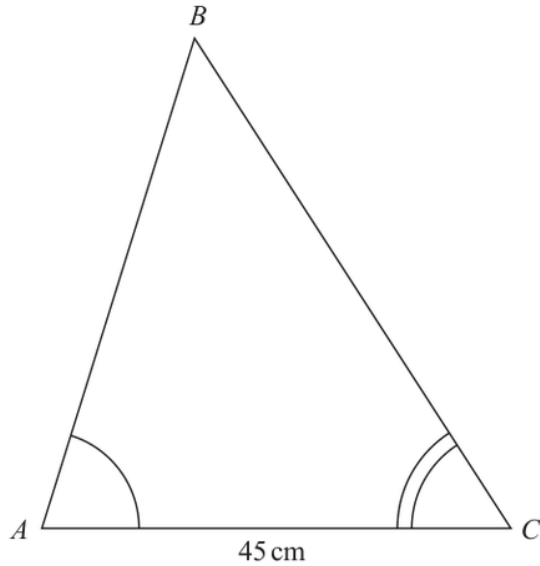
$$4(6) + 12(14) + 20(19) + 28(25) + 36(6) = 1488$$

$$\text{estimated mean} = \frac{1488}{70} = 21.257 \dots$$

FINAL ANSWER $\frac{31}{70}$, estimated mean 21.3 euros.

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Congruence, Similarity & Geometrical Proof**

Nov 2021 P2H - Nov 2021 | P2H

4 ABC and DEF are similar triangles.Diagram **NOT**
accurately drawn(a) Work out the length of AB cm
(2)Given that $BC = 54\text{ cm}$,(b) work out the length of EF cm
(2)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 4

TOPIC: **Congruence, Similarity & Geometrical Proof**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The scale factor from triangleThe scale factor from triangle DEF to triangle ABC is

$$\frac{AC}{DF} = \frac{45}{20} = 2.25$$

$$AB = DE \times 2.25 = 36 \times 2.25 = 81$$

2. Evaluate fractionGiven $BC = 54$,

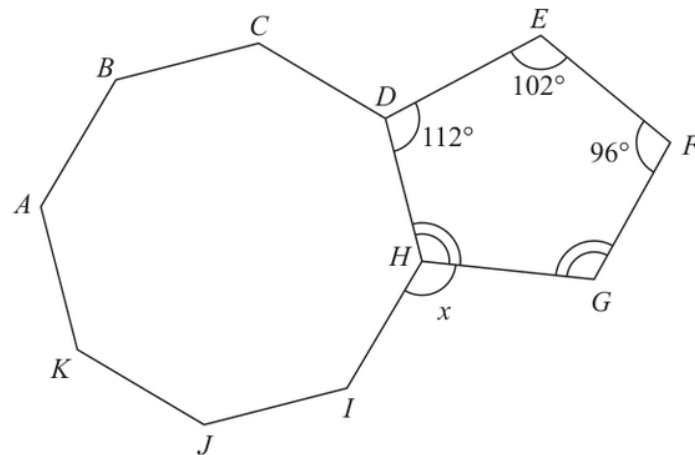
$$EF = \frac{54}{2.25} = 24$$

FINAL ANSWER $AB = 81$ cm, and $EF = 24$ cm.

PAST PAPER QUESTION**Q#: 5****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

Nov 2021 P2H - Nov 2021 | P2H

- 5 The diagram shows a regular octagon $ABCDHIJK$ and a pentagon $DEFGH$.

Diagram **NOT**
accurately drawnAngle $GHD = \text{angle } FGH$.

Work out the size of the angle marked x .
Show your working clearly.

o

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 5****TOPIC: Angles in Polygons & Parallel Lines**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the gradient**For pentagon $DEFGH$, the interior angle sum is

$$(5 - 2) \times 180 = 540^\circ$$

2. Let

Let

$$\angle GHD = \angle FGH = a$$

3. Find the gradient

Then

$$112 + 102 + 96 + a + a = 540$$

$$2a = 230$$

$$a = 115^\circ$$

4. A regular octagon has interior

A regular octagon has interior angle

$$\frac{(8 - 2) \times 180}{8} = 135^\circ$$

5. Find the gradientAt H , angles around a point add to 360° :

$$x = 360^\circ - 135^\circ - 115^\circ = 110^\circ$$

FINAL ANSWER

110°.

PAST PAPER QUESTION**Q#: 6****Marks: 3**TOPIC: **Percentages**

Nov 2021 P2H - Nov 2021 | P2H

- 6** Victor buys 12 bottles of apple juice for a total cost of \$21
Victor sells all 12 bottles at \$2.45 each bottle.

Work out Victor's percentage profit.

..... %

(Total for Question 6 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 6****Marks: 3**TOPIC: **Percentages**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Total selling price**

Total selling price:

$$12 \times 2.45 = 29.40$$

2. Profit

Profit:

$$29.40 - 21 = 8.40$$

3. Percentage profit

Percentage profit:

$$\frac{8.40}{21} \times 100 = 40$$

FINAL ANSWER

40%

PAST PAPER QUESTION**Q#: 7****Marks: 4****TOPIC: Compound Interest & Depreciation**

Nov 2021 P2H - Nov 2021 | P2H

- 7 Ali and Badia each have 25 000 dollars to invest.

Cyclone Bank	Tornado Bank
Invest 25 000 dollars 4.5% compound interest per year for 3 years	Invest 25 000 dollars Receive 1150 dollars interest each year for 3 years

Ali invests in the Cyclone Bank for 3 years.

Badia invests in the Tornado Bank for 3 years.

By the end of the 3 years, Ali will have received more interest than Badia.

How much more?

Show your working clearly.

Give your answer correct to the nearest dollar.

..... dollars

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 4****TOPIC: Compound Interest & Depreciation**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Ali's interest from Cyclone**

Ali's interest from Cyclone Bank is

$$25000(1.045)^3 - 25000 = 3529.153\dots$$

2. Badia's interest from Tornado

Badia's interest from Tornado Bank is

$$1150 \times 3 = 3450$$

3. Difference

Difference:

$$3529.153\dots - 3450 = 79.153\dots$$

4. Use compound interest

Correct to the nearest dollar,

$$79.153\dots \approx 79$$

FINAL ANSWER**\$79**

PAST PAPER QUESTION**Q#: 8****Marks: 4****TOPIC: Factorising**

Nov 2021 P2H - Nov 2021 | P2H

8 (a) Simplify $(3x^2y)^0$
(1)(b) (i) Factorise $x^2 - 5x - 36$
(2)(ii) Hence solve $x^2 - 5x - 36 = 0$
(1)

(Total for Question 8 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 8****Marks: 4****TOPIC: Factorising**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a) $(3x^2y)^0 = 1.$

2. (b)(i)

(b)(i)

$$x^2 - 5x - 36 = (x - 9)(x + 4)$$

3. Factorise expression(b)(ii) $x = 9$ or $x = -4.$ **FINAL ANSWER**1, $(x - 9)(x + 4)$, $x = 9$ or $x = -4$

PAST PAPER QUESTION**Q#: 9** **Marks: 5****TOPIC: Powers, Roots & Standard Form**

Nov 2021 P2H - Nov 2021 | P2H

- 9 A rainwater tank contains 2.4×10^7 raindrops.
The rainwater tank also contains 1.75×10^6 bacteria.
- (a) Work out the number of bacteria per raindrop in the tank.
Give your answer in standard form correct to 2 significant figures.

.....
(3)

A drop of rainwater contains 5.01×10^{21} atoms.

In a drop of rainwater the number of atoms is 3 times the number of molecules.

- (b) Work out the number of molecules in the rainwater tank.
Give your answer in standard form correct to one significant figure.

..... molecules
(2)

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 5****TOPIC: Powers, Roots & Standard Form**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert standard form**

For part (a), divide the number of bacteria by the number of raindrops:

$$\frac{1.75 \times 10^6}{2.4 \times 10^7} = \frac{1.75}{2.4} \times 10^{-1}$$

$$= 0.072916 \dots$$

2. Correct to 2 significant figures

Correct to 2 significant figures:

$$7.3 \times 10^{-2}$$

3. Convert standard formFor part (b), one drop contains 5.01×10^{21} atoms.**4. Convert standard form**

The number of molecules in one drop is

$$\frac{5.01 \times 10^{21}}{3} = 1.67 \times 10^{21}$$

5. Convert standard formThere are 2.4×10^7 drops, so the number of molecules in the tank is

$$(1.67 \times 10^{21})(2.4 \times 10^7) = 4.008 \times 10^{28}$$

6. Correct to one significant figure

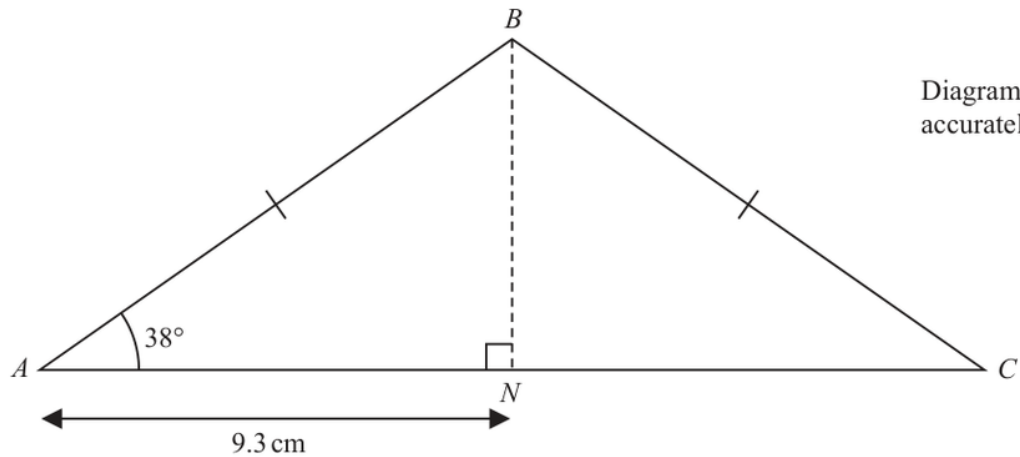
Correct to one significant figure:

$$4 \times 10^{28}$$

FINAL ANSWER(a) 7.3×10^{-2} , (b) 4×10^{28} molecules

PAST PAPER QUESTION**Q#: 10****Marks: 4****TOPIC: Area & Perimeter**

Nov 2021 P2H - Nov 2021 | P2H

10 ABC is an isosceles triangle with $BA = BC$.Diagram **NOT**
accurately drawn N is the point on AC such that $AN = 9.3$ cm and BN is perpendicular to AC .Work out the perimeter of triangle ABC .

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 10 is 4 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 4****TOPIC: Area & Perimeter**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**In right-angled triangle ABN ,

$$\cos 38^\circ = \frac{9.3}{AB}$$

$$AB = \frac{9.3}{\cos 38^\circ} = 11.802 \dots$$

2. Calculate areaSince ABC is isosceles,

$$BC = AB$$

3. Calculate area

Also,

$$AC = 2 \times 9.3 = 18.6$$

4. Perimeter

Perimeter:

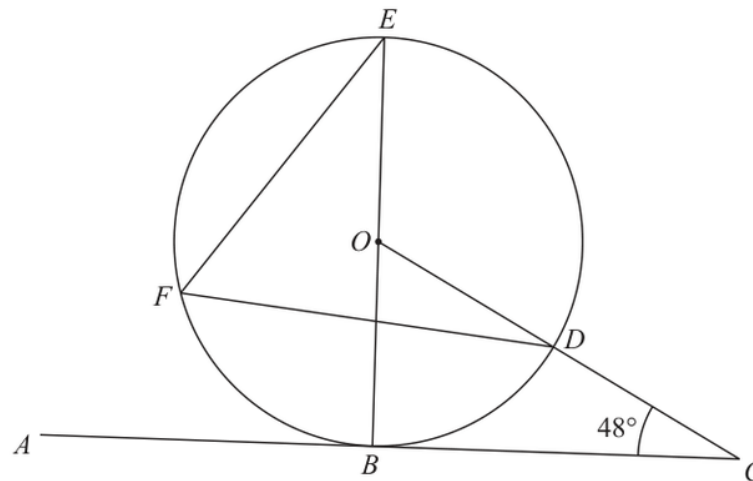
$$11.802 \dots + 11.802 \dots + 18.6 = 42.204 \dots$$

FINAL ANSWER

42.2 cm.

PAST PAPER QUESTION**Q#: 11****Marks: 3****TOPIC: Circle Theorems**

Nov 2021 P2H - Nov 2021 | P2H

11Diagram **NOT**
accurately drawn

B, D, E and F are points on a circle, centre O .
 ABC is a tangent to the circle.
 ODC is a straight line.

BOE is a diameter of the circle.

Angle $BCD = 48^\circ$

Find the size of angle DFE .

.....
 (Total for Question 11 is 3 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 3****TOPIC: Circle Theorems**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The tangent at is perpendicular**

The tangent at B is perpendicular to the radius OB , so

$$\angle OBC = 90^\circ$$

2. Use trigonometry

In triangle OBC ,

$$\angle BOC = 180^\circ - 90^\circ - 48^\circ = 42^\circ$$

3. Use trigonometry

Since O, D, C are on a straight line,

$$\angle BOD = 42^\circ$$

4. Use trigonometry

BOK is a diameter, so BOE is a straight line and

$$\angle DOE = 180^\circ - 42^\circ = 138^\circ$$

5. The angle at the circumference

The angle at the circumference is half the angle at the centre:

$$\angle DFE = \frac{138^\circ}{2} = 69^\circ$$

FINAL ANSWER

69° .

PAST PAPER QUESTION**Q#: 12****Marks: 7****TOPIC: Expanding Brackets**

Nov 2021 P2H - Nov 2021 | P2H

12 (a) Simplify $(64p^9q^{12})^{\frac{2}{3}}$
(2)(b) Write as a single fraction $\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x}$
Give your answer in its simplest form......
(2)(c) Expand and simplify $4x(x - 5)(2x + 3)$
Show your working clearly......
(3)**(Total for Question 12 is 7 marks)**

PAST PAPER SOLUTION**Q#: 12****Marks: 7****TOPIC: Expanding Brackets**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$(64p^9q^{12})^{2/3} = 16p^6q^8$$

2. (b)

(b)

$$\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x} = \frac{20 + 24 - 27}{30x} = \frac{17}{30x}$$

3. (c)

(c)

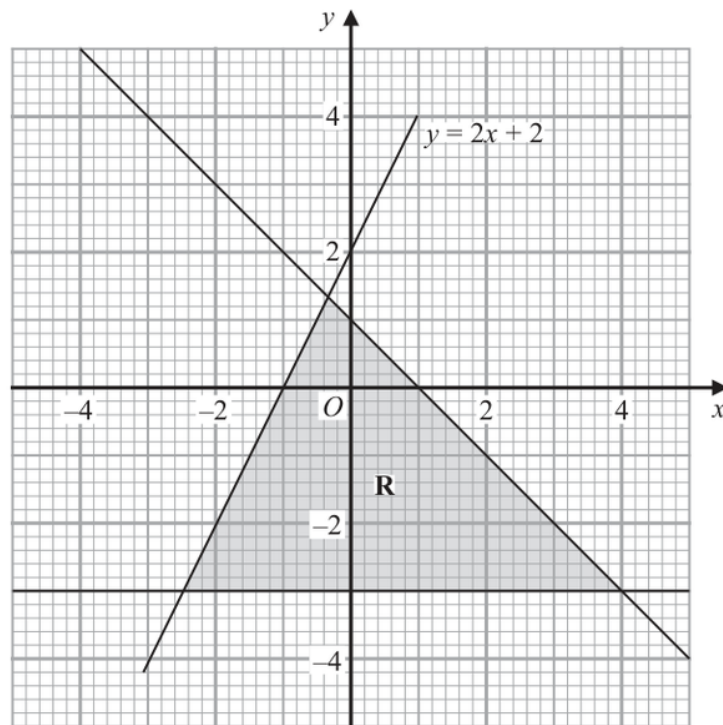
$$4x(x - 5)(2x + 3) = 4x(2x^2 - 7x - 15)$$

$$= 8x^3 - 28x^2 - 60x$$

FINAL ANSWER(a) $16p^6q^8$, (b) $\frac{17}{30x}$, (c) $8x^3 - 28x^2 - 60x$

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Graphing Inequalities**

Nov 2021 P2H - Nov 2021 | P2H

13

The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Write down the three inequalities that define **R**.

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 3****TOPIC: Graphing Inequalities**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The bottom boundary is so**

The bottom boundary is

$$y = -3$$

so

$$y \geq -3$$

2. The rising boundary is so

The rising boundary is

$$y = 2x + 2$$

so the region has

$$y \leq 2x + 2$$

3. Calculate value

The falling boundary goes through (0, 1) and (1, 0), so

$$x + y = 1$$

4. The region is below this

The region is below this line:

$$x + y \leq 1$$

FINAL ANSWER

$$y \geq -3, y \leq 2x + 2, x + y \leq 1.$$

PAST PAPER QUESTION**Q#: 14****Marks: 5****TOPIC: Histograms**

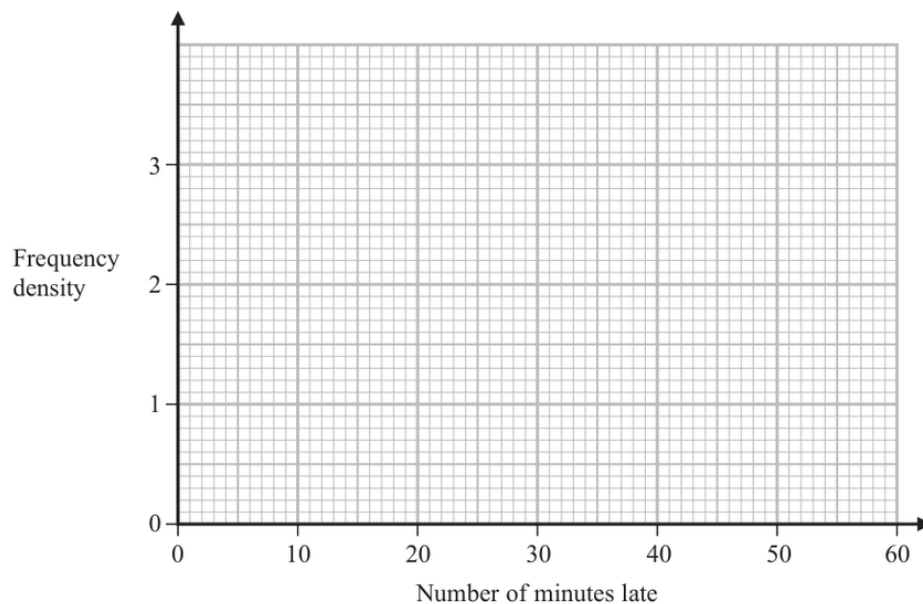
Nov 2021 P2H - Nov 2021 | P2H

14 Manuel collected information about the flights that arrived late at an airport last month.

The table gives information about the number of minutes that these flights were late.

Minutes late (L minutes)	Frequency
$0 < L \leq 10$	8
$10 < L \leq 15$	13
$15 < L \leq 25$	19
$25 < L \leq 40$	24
$40 < L \leq 60$	6

(a) On the grid, draw a histogram for this information.



(3)

Manuel selected at random a flight that was late by 25 minutes or less from his results.

(b) Work out an estimate for the probability that this flight was late by 5 minutes or less.

(2)

(Total for Question 14 is 5 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 5****TOPIC: Histograms**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The histogram densities are**

The histogram densities are

$$0 < L \leq 10 : \frac{8}{10} = 0.8$$

$$10 < L \leq 15 : \frac{13}{5} = 2.6$$

$$15 < L \leq 25 : \frac{19}{10} = 1.9$$

$$25 < L \leq 40 : \frac{24}{15} = 1.6$$

$$40 < L \leq 60 : \frac{6}{20} = 0.3$$

2. Use histogramGiven $L \leq 25$, the total frequency is

$$8 + 13 + 19 = 40$$

3. Use histogramFor $L \leq 5$, estimate half of the first class:

$$5(0.8) = 4$$

$$P(L \leq 5 \mid L \leq 25) = \frac{4}{40} = \frac{1}{10}$$

FINAL ANSWERhistogram densities 0.8, 2.6, 1.9, 1.6, 0.3, and probability $\frac{1}{10}$.

PAST PAPER QUESTION**Q#: 15****Marks: 6****TOPIC: Functions**

Nov 2021 P2H - Nov 2021 | P2H

15 The functions f and g are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of x that cannot be included in any domain of g
(1)(b) Find $gf(x)$

Simplify your answer.

$$gf(x) = \dots\dots\dots$$
(2)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$$g^{-1}(x) = \dots\dots\dots$$
(3)

(Total for Question 15 is 6 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 6****TOPIC: Functions**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$f(x) = 2x - 3, \quad g(x) = \frac{x}{3x + 1}$$

2. (a) The denominator of cannot(a) The denominator of g cannot be zero:

$$3x + 1 = 0$$

$$x = -\frac{1}{3}$$

3. (b)

(b)

$$gf(x) = g(f(x))$$

$$gf(x) = \frac{2x - 3}{3(2x - 3) + 1}$$

$$gf(x) = \frac{2x - 3}{6x - 8}$$

4. (c) Let

(c) Let

$$y = \frac{x}{3x + 1}$$

$$y(3x + 1) = x$$

$$3xy + y = x$$

$$x(3y - 1) = -y$$

$$x = \frac{y}{1 - 3y}$$

5. Evaluate fraction

So

$$g^{-1}(x) = \frac{x}{1 - 3x}$$

FINAL ANSWER

Excluded value $-\frac{1}{3}$, $gf(x) = \frac{2x-3}{6x-8}$, and $g^{-1}(x) = \frac{x}{1-3x}$.

EA

PAST PAPER QUESTION

Q#: 16

Marks: 3

TOPIC: **Combined & Conditional Probability**

Nov 2021 P2H - Nov 2021 | P2H

16 A box contains 15 counters.

There are 4 red counters, 5 green counters and the rest are yellow counters.

Niklas takes at random a counter from the box and writes down the colour of his counter.
He then puts the counter back into the box.

Sasha then takes at random a counter from the box and writes down the colour of her counter.

Work out the probability that the counters taken by Niklas and Sasha both have the same colour.

(Total for Question 16 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 16****Marks: 3****TOPIC: Combined & Conditional Probability**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**

There are 4 red, 5 green and 6 yellow counters.

2. Calculate probability

The first counter is replaced, so the probabilities are the same for Niklas and Sasha.

$$\begin{aligned}
 P(\text{same colour}) &= \left(\frac{4}{15}\right)^2 + \left(\frac{5}{15}\right)^2 + \left(\frac{6}{15}\right)^2 \\
 &= \frac{16 + 25 + 36}{225} \\
 &= \frac{77}{225}
 \end{aligned}$$

FINAL ANSWER

$$\frac{77}{225}$$

PAST PAPER QUESTION

Q#: 17

Marks: 3

TOPIC: **Surds**

Nov 2021 P2H - Nov 2021 | P2H

17 Express $\frac{8}{\sqrt{5}-1}$ in the form $\sqrt{a} + b$ where a and b are integers.

Show each stage of your working clearly.

(Total for Question 17 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 3****TOPIC: Surds**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

$$\frac{8}{\sqrt{5}-1}$$

2. Rationalise the denominator

Rationalise the denominator:

$$\begin{aligned}\frac{8}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} &= \frac{8(\sqrt{5}+1)}{5-1} \\ &= 2(\sqrt{5}+1) = 2\sqrt{5}+2\end{aligned}$$

3. Simplify surdIn the form $\sqrt{a} + b$,

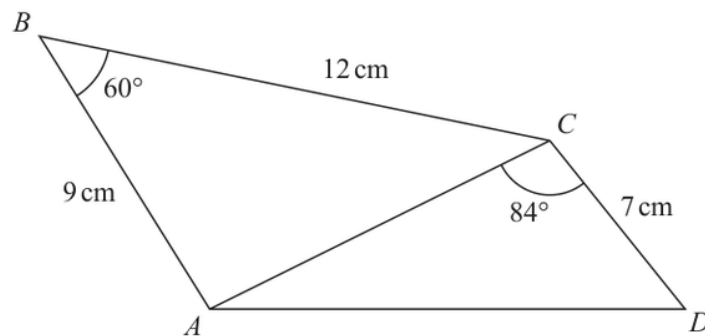
$$2\sqrt{5}+2 = \sqrt{20}+2$$

FINAL ANSWER

$$\sqrt{20}+2$$

PAST PAPER QUESTION**Q#: 18****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Nov 2021 P2H - Nov 2021 | P2H

18 Here is a quadrilateral $ABCD$.Diagram **NOT**
accurately drawn

Calculate the area of quadrilateral $ABCD$.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... cm^2

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Area of triangle**Area of triangle ABC :

$$\frac{1}{2}(9)(12) \sin 60^\circ = 46.765 \dots$$

2. Find using the cosine ruleFind AC using the cosine rule:

$$AC^2 = 9^2 + 12^2 - 2(9)(12) \cos 60^\circ$$

$$AC = 10.816 \dots$$

3. Area of triangleArea of triangle ACD :

$$\frac{1}{2}(10.816 \dots)(7) \sin 84^\circ = 37.642 \dots$$

4. Total area

Total area:

$$46.765 \dots + 37.642 \dots = 84.407 \dots$$

FINAL ANSWER84.4 cm².

PAST PAPER QUESTION**Q#: 19****Marks: 6****TOPIC: Coordinate Geometry**

Nov 2021 P2H - Nov 2021 | P2H

- 19** The straight line **L** has equation $x - y = 3$
The curve **C** has equation $3x^2 - y^2 + xy = 9$

L and **C** intersect at the points *P* and *Q*.

Find the coordinates of the midpoint of *PQ*.
Show clear algebraic working.

(..... ,)

(Total for Question 19 is 6 marks)

EA

PAST PAPER SOLUTION**Q#: 19****Marks: 6****TOPIC: Coordinate Geometry**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Line is**Line L is

$$x - y = 3$$

2. Calculate value

So

$$y = x - 3$$

3. The curve is

The curve is

$$3x - y + y^2 = 9$$

4. SubstituteSubstitute $y = x - 3$:

$$3x - (x - 3) + (x - 3)^2 = 9$$

$$2x + 3 + x^2 - 6x + 9 = 9$$

$$x^2 - 4x + 3 = 0$$

$$(x - 1)(x - 3) = 0$$

5. Calculate valueSo $x = 1$ or $x = 3$.**6. Calculate value**If $x = 1$, then $y = -2$. If $x = 3$, then $y = 0$.**7. The midpoint of and is**The midpoint of $(1, -2)$ and $(3, 0)$ is

$$\left(\frac{1+3}{2}, \frac{-2+0}{2} \right) = (2, -1)$$

FINAL ANSWER $(2, -1)$.

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Sequences**

Nov 2021 P2H - Nov 2021 | P2H

20 Here are the first four terms of an arithmetic series.

$$k \quad \frac{3k}{4} \quad \frac{k}{2} \quad \frac{k}{4}$$

Given that the 15th term of the series is $(90 + 2k)$,

calculate the sum of the first 30 terms of the series.

(Total for Question 20 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 5**TOPIC: **Sequences**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The first four terms are**

The first four terms are

$$k, \quad \frac{3k}{4}, \quad \frac{k}{2}, \quad \frac{k}{4}$$

2. The common difference is

The common difference is

$$d = -\frac{k}{4}$$

3. The 15th term is

The 15th term is

$$u_{15} = k + 14 \left(-\frac{k}{4} \right)$$

$$u_{15} = k - \frac{7k}{2} = -\frac{5k}{2}$$

4. Given

Given

$$-\frac{5k}{2} = 90 + 2k$$

$$-5k = 180 + 4k$$

$$k = -20$$

5. Evaluate fraction

So

$$a = -20, \quad d = 5$$

$$S_{30} = \frac{30}{2}(2(-20) + 29(5))$$

$$S_{30} = 15(105) = 1575$$

FINAL ANSWER

1575

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Sequences**

Nov 2021 P2H - Nov 2021 | P2H

20 Here are the first four terms of an arithmetic series.

$$k \quad \frac{3k}{4} \quad \frac{k}{2} \quad \frac{k}{4}$$

Given that the 15th term of the series is $(90 + 2k)$,

calculate the sum of the first 30 terms of the series.

(Total for Question 20 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 5**TOPIC: **Sequences**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The first four terms are**

The first four terms are

$$k, \quad \frac{3k}{4}, \quad \frac{k}{2}, \quad \frac{k}{4}$$

2. The common difference is

The common difference is

$$d = -\frac{k}{4}$$

3. The 15th term is

The 15th term is

$$u_{15} = k + 14 \left(-\frac{k}{4} \right)$$

$$u_{15} = k - \frac{7k}{2} = -\frac{5k}{2}$$

4. Given

Given

$$-\frac{5k}{2} = 90 + 2k$$

$$-5k = 180 + 4k$$

$$k = -20$$

5. Evaluate fraction

So

$$a = -20, \quad d = 5$$

$$S_{30} = \frac{30}{2}(2(-20) + 29(5))$$

$$S_{30} = 15(105) = 1575$$

FINAL ANSWER

1575

PAST PAPER QUESTION**Q#: 21** **Marks: 6****TOPIC: Transformations of Graphs**

Nov 2021 P2H - Nov 2021 | P2H

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

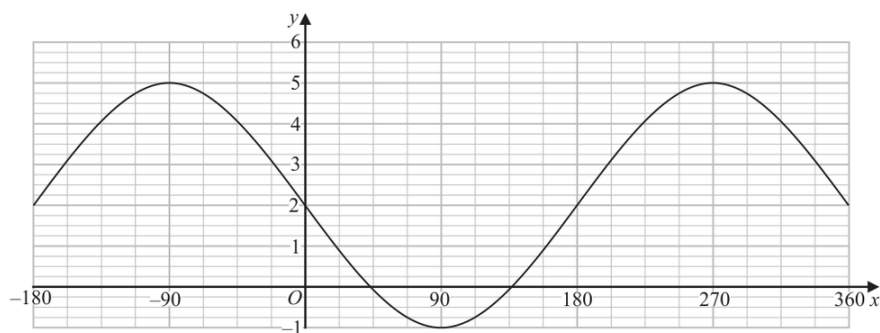
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(3)

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 6****TOPIC: Transformations of Graphs**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$f(x) = 9 - 3(x + 2)^2$$

2. This is in completed square

This is in completed square form, so the maximum point is

$$(-2, 9)$$

3. Use matrices

For the translation

$$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

4. the graph moves 4 units

the graph moves 4 units right. Replace x by $x - 4$:

$$y = 9 - 3((x - 4) + 2)^2$$

$$y = 9 - 3(x - 2)^2$$

5. The curve has equation

The curve T has equation

$$y = 3(x + 2)^2 - 9$$

6. This is

This is

$$y = -\left(9 - 3(x + 2)^2\right)$$

7. Apply transformation

So C is reflected in the x -axis.

8. Use trigonometry

For the cosine graph, the maximum is 5 and the minimum is -1 .

$$a = \frac{5 - (-1)}{2} = 3$$

$$c = \frac{5 + (-1)}{2} = 2$$

9. Use trigonometry

For

$$y = a \cos(x - b)^\circ + c$$

10. Read the graph

a maximum occurs when $x - b = 0$. The graph has a maximum at $x = 270^\circ$, so

$$b = 270$$

FINAL ANSWER

$A = (-2, 9)$, $y = 9 - 3(x - 2)^2$, reflection in the x -axis, $a = 3$, $b = 270$, $c = 2$

EA

PAST PAPER QUESTION**Q#: 21****Marks: 6****TOPIC: Transformations of Graphs**

Nov 2021 P2H - Nov 2021 | P2H

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

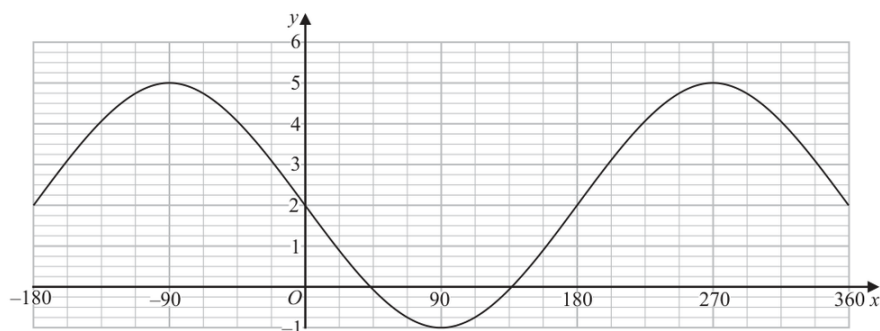
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(3)

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 6****TOPIC: Transformations of Graphs**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$f(x) = 9 - 3(x + 2)^2$$

2. This is in completed square

This is in completed square form, so the maximum point is

$$(-2, 9)$$

3. Use matrices

For the translation

$$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

4. the graph moves 4 units

the graph moves 4 units right. Replace x by $x - 4$:

$$y = 9 - 3((x - 4) + 2)^2$$

$$y = 9 - 3(x - 2)^2$$

5. The curve has equation

The curve T has equation

$$y = 3(x + 2)^2 - 9$$

6. This is

This is

$$y = -\left(9 - 3(x + 2)^2\right)$$

7. Apply transformation

So C is reflected in the x -axis.

8. Use trigonometry

For the cosine graph, the maximum is 5 and the minimum is -1 .

$$a = \frac{5 - (-1)}{2} = 3$$

$$c = \frac{5 + (-1)}{2} = 2$$

9. Use trigonometry

For

$$y = a \cos(x - b)^\circ + c$$

10. Read the graph

a maximum occurs when $x - b = 0$. The graph has a maximum at $x = 270^\circ$, so

$$b = 270$$

FINAL ANSWER

$A = (-2, 9)$, $y = 9 - 3(x - 2)^2$, reflection in the x -axis, $a = 3$, $b = 270$, $c = 2$

EA

PAST PAPER QUESTION

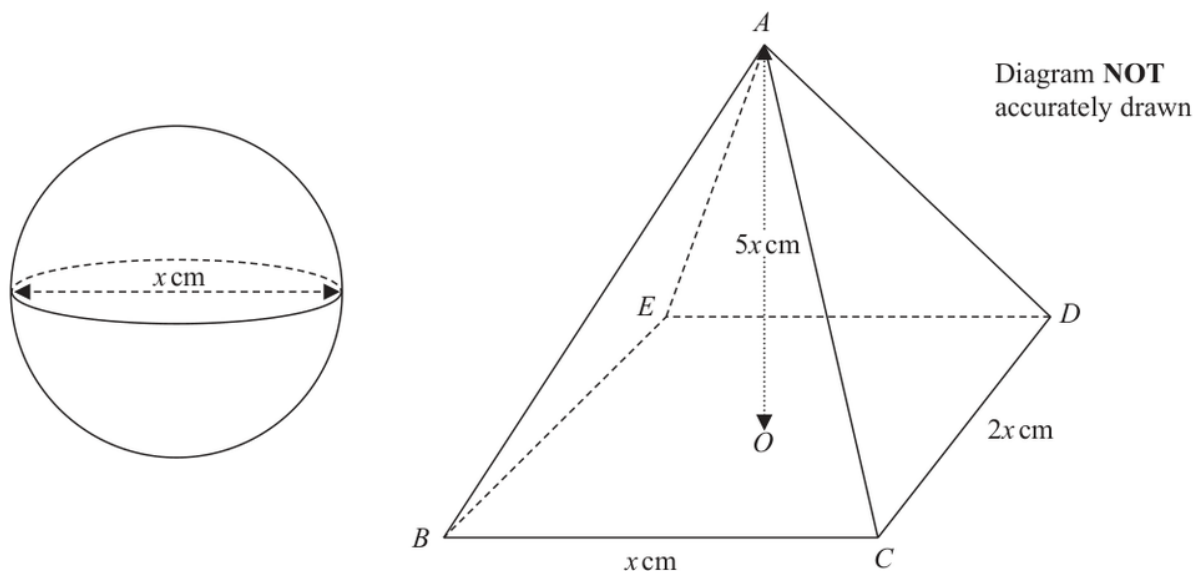
Q#: 22

Marks: 6

TOPIC: **Volume & Surface Area**

Nov 2021 P2H - Nov 2021 | P2H

- 22 The diagram shows a sphere of diameter x cm and a pyramid $ABCDE$ with a horizontal rectangular base $BCDE$.



The vertex A of the pyramid is vertically above the centre O of the base so that $AB = AC = AD = AE$.

$BC = x$ cm, $CD = 2x$ cm and $AO = 5x$ cm.

The volume of the sphere is $288\pi \text{ cm}^3$

Calculate the total surface area of the pyramid.
Give your answer correct to the nearest cm^2

..... cm^2

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Volume & Surface Area**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**The sphere has diameter x , so its radius is

$$\frac{x}{2}$$

2. The volume of the sphereThe volume of the sphere is $288\pi \text{ cm}^3$:

$$\frac{4}{3}\pi \left(\frac{x}{2}\right)^3 = 288\pi$$

$$\frac{\pi x^3}{6} = 288\pi$$

$$x^3 = 1728$$

$$x = 12$$

3. The rectangular base of theThe rectangular base of the pyramid has dimensions x and $2x$, so its area is

$$x(2x) = 2x^2$$

4. Use trigonometryFor the two faces with base x , the slant height is

$$\sqrt{(5x)^2 + x^2} = x\sqrt{26}$$

5. Their total area is

Their total area is

$$2\left(\frac{1}{2}x \cdot x\sqrt{26}\right) = x^2\sqrt{26}$$

6. Use trigonometryFor the two faces with base $2x$, the slant height is

$$\sqrt{(5x)^2 + \left(\frac{x}{2}\right)^2} = \frac{x\sqrt{101}}{2}$$

7. Their total area is

Their total area is

$$2 \left(\frac{1}{2}(2x) \cdot \frac{x\sqrt{101}}{2} \right) = x^2\sqrt{101}$$

8. Total surface area

Total surface area:

$$2x^2 + x^2\sqrt{26} + x^2\sqrt{101}$$

9. SubstituteSubstitute $x = 12$:

$$144(2 + \sqrt{26} + \sqrt{101}) = 2469.440\dots$$

10. Correct to the nearestCorrect to the nearest cm^2 :

$$2469$$

FINAL ANSWER

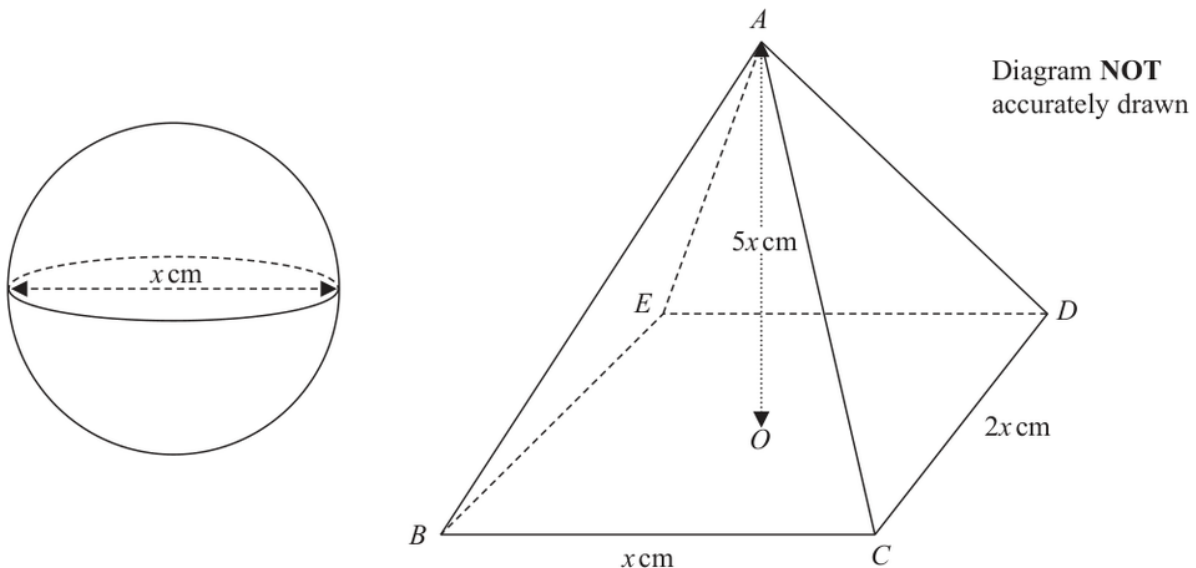
$$2469 \text{ cm}^2$$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Volume & Surface Area**

Nov 2021 P2H - Nov 2021 | P2H

- 22** The diagram shows a sphere of diameter x cm and a pyramid $ABCDE$ with a horizontal rectangular base $BCDE$.



The vertex A of the pyramid is vertically above the centre O of the base so that $AB = AC = AD = AE$.

$BC = x$ cm, $CD = 2x$ cm and $AO = 5x$ cm.

The volume of the sphere is $288\pi \text{ cm}^3$

Calculate the total surface area of the pyramid.
Give your answer correct to the nearest cm^2

..... cm^2

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Volume & Surface Area**

Nov 2021 P2H - Nov 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**The sphere has diameter x , so its radius is

$$\frac{x}{2}$$

2. The volume of the sphereThe volume of the sphere is $288\pi \text{ cm}^3$:

$$\frac{4}{3}\pi \left(\frac{x}{2}\right)^3 = 288\pi$$

$$\frac{\pi x^3}{6} = 288\pi$$

$$x^3 = 1728$$

$$x = 12$$

3. The rectangular base of theThe rectangular base of the pyramid has dimensions x and $2x$, so its area is

$$x(2x) = 2x^2$$

4. Use trigonometryFor the two faces with base x , the slant height is

$$\sqrt{(5x)^2 + x^2} = x\sqrt{26}$$

5. Their total area is

Their total area is

$$2\left(\frac{1}{2}x \cdot x\sqrt{26}\right) = x^2\sqrt{26}$$

6. Use trigonometryFor the two faces with base $2x$, the slant height is

$$\sqrt{(5x)^2 + \left(\frac{x}{2}\right)^2} = \frac{x\sqrt{101}}{2}$$

7. Their total area is

Their total area is

$$2 \left(\frac{1}{2}(2x) \cdot \frac{x\sqrt{101}}{2} \right) = x^2\sqrt{101}$$

8. Total surface area

Total surface area:

$$2x^2 + x^2\sqrt{26} + x^2\sqrt{101}$$

9. Substitute

Substitute $x = 12$:

$$144(2 + \sqrt{26} + \sqrt{101}) = 2469.440 \dots$$

10. Correct to the nearest

Correct to the nearest cm^2 :

$$2469$$

FINAL ANSWER

$$2469 \text{ cm}^2$$

EA